

tf.compat.v1.linalg.trace Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/linalg/trace

Compute the trace of a tensor x .

Function Signatures

```
tf.compat.v1.linalg.trace( x, name=None )  
trace(x)
```

Code Examples

```
tf.compat.v1.linalg.trace(  
x, name=None  
)
```

```
tf.compat.v1.linalg.trace(  
x, name=None  
)
```

```
x = tf.constant([[1, 2], [3, 4]])
tf.linalg.trace(x) # 5
x = tf.constant([[1, 2, 3],
[4, 5, 6],
[7, 8, 9]])
tf.linalg.trace(x) # 15
x = tf.constant([[[1, 2, 3],
[4, 5, 6],
[7, 8, 9]],
[[-1, -2, -3],
[-4, -5, -6],
[-7, -8, -9]]])
tf.linalg.trace(x) # [15, -15]
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x = tf.constant([[1, 2], [3, 4]])
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tf.linalg.trace(x) # [15, -15]
```

Documentation

Compute the trace of a tensor x .

`trace(x)` returns the sum along the main diagonal of each inner-most matrix in x . If x is of rank k with shape $[I, J, K, \dots, L, M, N]$, then output is a tensor of rank $k-2$ with dimensions $[I, J, K, \dots, L]$ where

$\text{output}[i, j, k, \dots, l] = \text{trace}(x[i, j, k, \dots, l, :, :])$

For example:

Returns